*Plant, Cell & Environment*

WILEY

Plant, Cell & Environment

TECHNICAL REPORT OPEN ACCESS

# Estimating Leaf $CO_2$ Assimilation in $C_3$ Plants Using a Handheld Porometer With Chlorophyll Fluorometer in Field Conditions

Kensuke Kimura[1] | Erina Fushimi[1] | Etsushi Kumagai[1] | Koichi Nomura[2] | Toshinori Matsunami[3] | Shohei Konno[4] | Atsushi Maruyama[1]

[1]Institute for Agro-Environmental Sciences, National Agriculture and Food Research Organization (NARO), Tsukuba, Japan | [2]IoP Collaborative Creation Center, Kochi University, Nankoku, Japan | [3]Tohoku Agricultural Research Center, National Agriculture and Food Research Organization (NARO), Morioka, Japan | [4]Institute of Fruit Tree and Tea Science, National Agriculture and Food Research Organization (NARO), Tsukuba, Japan

**Correspondence:** Kensuke Kimura (kimurak081@naro.affrc.go.jp; kimura.kensuke673@naro.go.jp)

**Received:** 28 November 2024 | **Revised:** 31 May 2025 | **Accepted:** 3 June 2025

**Funding:** This study was partially supported by the Japan Society for the Promotion of Science (JSPS) KAKENHI, under Grant Number 21K14950 and 25H00953. This study was also supported by a Cabinet Office grant in aid, the Advanced Next-Generation Greenhouse Horticulture by IoP (Internet of Plants), Japan, and by a Cabinet Office grant in aid, Evolution to Society 5.0 Agriculture Driven by IoP (Internet of Plants), Japan.

**Keywords:** chlorophyll fluorescence | gas exchange | herbaceous plants | high-throughput phenotyping | photosynthesis | pulse amplitude modulated (PAM) fluorometry | stomatal conductance | sunlight | woody plants

## ABSTRACT

Gas exchange measurement is the gold standard method for determining leaf $CO_2$ assimilation rate ($A_n$). However, conventional systems for measuring $A_n$ often require time and/or effort to collect numerous samples in the field owing to their high weight and large size. Here, we present an efficient and convenient method for estimating $A_n$ using a handheld porometer with a chlorophyll fluorometer, facilitating on-the-go assessment of $A_n$ in the field. This porometer-fluorometer method integrates the measured stomatal conductance and quantum yield of photochemistry in PSII into a biochemical photosynthesis model, incorporating model uncertainties into a single calibrated parameter. Using this method, we successfully estimated $A_n$ variations in 12 species under field conditions, with a root mean square error of 2.0 $\mu$mol $m^{-2}$ $s^{-1}$, despite using the common parameter set. In contrast, without calibration (i.e., with the often-assumed parameter value), this method greatly overestimated $A_n$. These results highlight the importance of appropriate calibration depending on prevailing conditions, particularly the light source. In summary, this method demonstrates the potential for accessible, high-throughput, and accurate estimation of $A_n$ in diverse plants, thereby addressing a key bottleneck in field-based phenotyping of photosynthesis. However, further studies are required to reduce the uncertainties imposed on the calibrated parameter.

## 1 | Introduction

Photosynthesis is essential for plant growth and development and provides critical insights into plant health and productivity. In agriculture, understanding photosynthetic performance is valuable for breeding crops and trees with enhanced photosynthetic efficiency and yield potential (Zhu et al. 2010; Simkin et al. 2019). Gas exchange measurement is the gold standard for measuring leaf photosynthesis (i.e., leaf $CO_2$ assimilation rate, $A_n$). Conventional systems using an assimilation chamber and infrared gas analyzers can provide detailed photosynthetic information, such as the maximum rates of RuBisCO carboxylation and electron transport, owing to the precise environmental control of these systems (Saathoff and Welles 2021). However, their large size and high weight (~10 kg) demand significant effort to hold them throughout the measurement, making them impractical for on-the-go field measurements, particularly when moving across multiple leaves in a large field. Although tripods can reduce the effort required to fix the equipment, setting up a tripod for each measurement is time-consuming. Moreover, gas exchange systems require stabilization time in minutes, raising the total time for each measurement to at least several minutes. Such effort and time considerations may make researchers, particularly who need to screen numerous samples in a large field (such as breeding programs in agriculture), refrain from gas exchange measurements and miss opportunities to phenotype photosynthesis. Therefore, there is a need for more accessible and high-throughput techniques than conventional systems.

Recently, a handheld porometer equipped with a pulse-amplitude modulated (PAM) chlorophyll fluorometer was developed (e.g., LI-600 from LI-COR bioscience), offering a practical tool for measuring critical photosynthetic parameters, including stomatal conductance ($g_s$) and quantum yield of photochemistry in PSII ($\Phi_{PSII}$). This compact and lightweight (~1 kg) device can simultaneously measure $g_s$ and $\Phi_{PSII}$ within seconds, making it well-suited for high-throughput field phenotyping and significantly easier to use for on-the-go assessment than traditional gas exchange systems. Although these instruments do not directly measure leaf $CO_2$ assimilation ($A_n$), He and Edwards (1996) proposed a mechanistic model that estimates $A_n$ using $g_s$ and $\Phi_{PSII}$ as inputs. This model was developed based on the biochemical processes of photosynthesis and is applicable to $C_3$ plants. Although the original model requires an empirical model to estimate $g_s$ (Leuning 1995), modern porometers can directly measure $g_s$; thus, a more accurate estimation of $A_n$ is expected (Kitao et al. 2021). Therefore, integrating handheld instruments with mechanistic models offers an accessible, high-throughput, and accurate approach to estimating $A_n$ in diverse $C_3$ species.

Despite the potential of this porometer-fluorometer approach, its practical application is limited by the uncertainties in estimating the electron transport rate ($J$) within the model. $J$ is often estimated using $\Phi_{PSII}$ measured by a chlorophyll fluorometer, as follows (Genty et al. 1989):

$$J = \Phi_{PSII}\, \mathrm{PPFD}\, \alpha\, \beta, \tag{1}$$

where PPFD is the photosynthetic photon flux density, $\alpha$ is the leaf absorptance of PPFD, and $\beta$ is the fraction of the absorbed light allocated to PSII. Typically, $\alpha$ and $\beta$ are set at 0.84 (or 0.85) and 0.5, respectively, although these values likely differ among species (Laisk and Loreto 1996) and growth light conditions (Chow et al. 1990; Murakami et al. 2018). Some studies have suggested additional parameters in Equation (1) to correct for $J$ overestimations due to light absorption by non-photosynthetic pigments (Loreto et al. 2009; McClain and Sharkey 2020), alternative electron sinks (Yin et al. 2009), and mesophyll layer heterogeneity (Evans 2009). These errors are remarkable under high blue-to-red light ratios, such as natural sunlight (McClain and Sharkey 2020), leading many studies to use artificial light with a minimal blue light to minimize $J$ overestimation. Consequently, few studies have examined $J$ estimation under sunlight, the most common light source for plant growth, leaving the necessary corrections across species and growth conditions unclear.

In this study, we tested whether combining a handheld porometer with a chlorophyll fluorometer and a biochemical photosynthesis model could accurately estimate $A_n$ variations under natural sunlight across 12 $C_3$ species grown in varied conditions. We assessed how empirical calibration affected estimation error correction and discussed the potential for field applications of this method. This approach offers a practical solution to the bottleneck in phenotyping photosynthesis under field conditions, enabling more accessible and accurate $A_n$ assessments in diverse environments.

## 2 | Materials and Methods

### 2.1 | Model for Estimating Leaf $CO_2$ Assimilation

The relationship between $A_n$ and $J$ was formulated based on the electron requirements for carboxylation and oxygenation using the biochemical photosynthesis model of Farquhar et al. (1980) as follows:

$$A_n = \frac{J\,(C_i - \Gamma^*)}{4(C_i + 2\Gamma^*)} - R_d, \tag{2}$$

where $C_i$ is the intercellular $CO_2$ concentration, $\Gamma^*$ is the $CO_2$ compensation point in the absence of day respiration, and $R_d$ is day respiration rate. Although $A_n$ is limited by either RuBisCo, ribulose 1,5-bisphosphate (RuBP) regeneration, or triose phosphate utilization (TPU) (Farquhar et al. 1980; Sharkey 1985), Equation (2) can be applied to all three limitation phases when "actual" $J$ is directly quantified (Harley et al. 1992). $J$ was quantified by modifying Equation (1) after Loreto et al. (2009), Yin et al. (2009), and van der Putten et al. (2018) are as follows:

$$J = \Phi_{PSII}\, \mathrm{PPFD}\, \alpha\, \beta\, \gamma \left(1 - \frac{f_{pseudo}}{1 - f_{cyc}}\right)\xi = \Phi_{PSII}\, \mathrm{PPFD}\, s, \tag{3}$$

where $\gamma$ is the correction factor accounting for the light absorption by non-photosynthetic pigments (Loreto et al. 2009; McClain and Sharkey 2020); $f_{pseudo}$ and $f_{cyc}$ are the fractions of total electron flux used as pseudo-cyclic and cyclic electron transport in PSI, respectively (Yin et al. 2009), representing alternative electron flows; and $\xi$ is the ratio of true $\Phi_{PSII}$ and measured $\Phi_{PSII}$ accounting for measurement errors of $\Phi_{PSII}$ (van der Putten et al. 2018) owing to mesophyll layer heterogeneity (Evans 2009). These parameters, in addition to $\alpha$ and $\beta$, are difficult to determine under field conditions; thus, we lumped these hard-to-measure parameters as the calibrated parameter $s$ (Yin et al. 2009; van der Putten et al. 2018).

$C_i$ was calculated by the $CO_2$ transfer of the leaf as follows:

$$C_i = C_a - \frac{A_n}{g_{tc}}, \tag{4}$$

where $C_a$ is the ambient air $CO_2$ concentration and $g_{tc}$ is the leaf total conductance for $CO_2$ transfer on both leaf sides. Porometers generally measure the conductance for one side of a leaf, and thus the measurement of the adaxial and abaxial sides of the leaf is required to calculate $g_{tc}$. According to the electrical circuit theory described by *Ohm's law*, the conductances of the leaf boundary layer and stomata are connected in series, whereas those for the adaxial and abaxial sides of the leaf are parallel. This results in the calculation of $g_{tc}$ as follows:

$$g_{tc} = \left(1.37g_{bw,ad}^{-1} + 1.6\,g_{sw,ad}^{-1}\right)^{-1} + \left(1.37g_{bw,ab}^{-1} + 1.6\,g_{sw,ab}^{-1}\right)^{-1}, \tag{5}$$

where $g_{bw}$ and $g_{sw}$ are the conductance of the leaf boundary layer and stomata for water vapor transfer, respectively, and the subscripts ad and ab represent the adaxial and abaxial sides of the leaf, respectively. The coefficients 1.37 and 1.6 are the conversion factors from water vapor to $CO_2$ diffusivity.

$\Gamma^*$ was determined as follows:

$$\Gamma^* = \frac{0.5\,O}{S_{c/o}}, \tag{6}$$

where $O$ is the oxygen concentration (210 mmol mol$^{-1}$ in this study), $S_{c/o}$ is the RuBisCO specificity factor, and the temperature response of $S_{c/o}$ was determined using the Arrhenius equation, as follows:

$$S_{c/o} = S_{c/o,25}\exp\left[\frac{E_{a,S_{c/o}}(T_{l,K} - 298.15)}{298.15\,R\,T_{l,K}}\right], \tag{7}$$

where $S_{c/o,25}$ is $S_{c/o}$ at 25°C, $E_{a,S_{c/o}}$ is the activation energy of $S_{c/o}$ to temperature, $R$ is the universal gas constant (8.314 J mol$^{-1}$ K$^{-1}$), $T_l$ is the leaf temperature, and the subscript K represents kelvin unit.

$R_d$ was determined using the Arrhenius equation, as follows:

$$R_d = R_{d,25}\exp\left[\frac{E_{a,R_d}(T_{l,K} - 298.15)}{298.15\,R\,T_{l,K}}\right], \tag{8}$$

and $R_{d,25}$ is $R_d$ at 25°C, $E_{a,R_d}$ is the activation energy of $R_d$ to temperature.

$A_n$ can be estimated by Equations (2) to (8) using measurable variables of $\Phi_{PSII}$, PPFD, $g_{bw}$, $g_{sw}$, $C_a$, and $T_l$ along with parameters of $S_{c/o,25}$, $E_{a,S_{c/o}}$, $R_{d,25}$, $E_{a,R_d}$, and $s$. The analytical solution for $A_n$ is described in Supporting Information S1: Note S1, and a template file for $A_n$ estimation is provided in Supporting Information S2. In this study, $S_{c/o,25}$ and $E_{a,S_{c/o}}$ were set to 2.57 mmol mol$^{-1}$ and –28550 J mol$^{-1}$, respectively, which were re-calculated based on the In Vitro average values of $C_3$ plants (Galmés et al. 2016). $R_{d,25}$ and $E_{a,R_d}$ were set to 1.27 µmol m$^{-2}$ s$^{-1}$ and 59170 J mol$^{-1}$, based on the average values of wheat under photorespiratory conditions (Fang et al. 2022). Thus, $s$ was the sole calibrated parameter in this method, and uncertainties in other parameters, as well as those not explained by this model (e.g., mesophyll conductance, $g_m$), were imposed on the calibrated $s$. Although $s$ is generally calibrated under low $O_2$ conditions to remove the influence of photorespiration (i.e., $C_i$ and $\Gamma^*$), this cannot calibrate the uncertainties of $C_i$ and $\Gamma^*$ and requires further calibration to accurately estimate $A_n$. Moreover, changing the $O_2$ level is time-consuming in the field. Therefore, we calibrated $s$ under ambient $O_2$ conditions to avoid multiple calibrations and to maintain the high-throughput capability of the proposed method for field applications. The detailed processes of the calibration are described in Section 2.5.

## 2.2 | Plant Materials

We selected six herbaceous and six woody plants that were grown under sunlight at different experimental sites in Japan. Among the six herbaceous species, wheat (*Triticum aestivum*) and soybean (*Glycine max*) were grown outdoors, whereas tomato (*Solanum lycopersicum*), eggplant (*Solanum melongena*), bell pepper (*Capsicum annuum*), and spinach (*Spinacia oleracea*) were grown in greenhouses. All six woody species [apple (*Malus domestica*), peach (*Prunus persica*), pear (*Pyrus pyrifolia*), loquat (*Rhaphiolepis bibas*), lemon (*Citrus limon*), and orange (*Citrus unshiu*)] were grown outdoors. We used three cultivars of soybean and one cultivar for each of the other species; thus, 14 cultivars from the 12 species were used in this study. The detailed growth conditions are described in Supporting Information S1: Note S2.

## 2.3 | Measurements of Leaf Gas Exchange and Chlorophyll Fluorescence

To test the porometer-fluorometer method, gas exchange and chlorophyll fluorescence of leaves were measured in the field under sunlight using a gas exchange measurement system (LI-6800; Li-Cor, USA) and a porometer equipped with a chlorophyll fluorometer (LI-600; Li-Cor, USA). During the day, $A_n$ was measured on a fully expanded leaf using the LI-6800 with a clear top chamber to capture natural sunlight, and with a $1 \times 3$ cm$^2$ chamber aperture plate to clamp thin leaves and avoid heterogeneous distribution of $A_n$ on the measurement area as much as possible. After clamping the leaf with the adaxial side facing up, the measurement area faced toward the sun by changing the angle of the chamber head to avoid partial shading of the measurement area. During the measurement, the air temperature ($T_a$) and relative humidity ($h$) in the LI-6800 system were maintained at values monitored on the LI-600 porometer to match the environmental conditions between the instruments. $g_{bw}$ in the LI-6800 system was maintained at approximately 2.9 $H_2O$ mol m$^{-2}$ s$^{-1}$, which is equivalent to the value in the LI-600 porometer, by adjusting the fan speed in the chamber. $C_a$ in the LI-6800 system was also maintained at ambient values measured using a nondispersive infrared sensor (TR-76Ui, T&D, Japan). Other commercialized porometers (e.g., MINI-PAM-II/POROMETER; WALZ) can simultaneously measure $C_a$ and perform an estimation independently. The data were logged once $A_n$ and $g_{sw}$ in the LI-6800 system stabilized.

Immediately after removing the leaf from the chamber, $g_{sw}$, $g_{bw}$, PPFD, $\Phi_{PSII}$, and $T_l$ were measured using the LI-600 porometer equipped with the fluorometer. A multiphase flash method (Loriaux et al. 2013) was used to measure $\Phi_{PSII}$. The measurement was first performed on the adaxial side of the leaf, with PPFD values matching those recorded in the LI-6800 system, and repeated three times along the measurement area of $A_n$. The same procedure was then conducted on the abaxial side while maintaining the angle and orientation of the leaf (i.e., with the LI-600 porometer upside down). These measurements were repeated for different leaves of each species and cultivar over 21 days (January to November 2024). The procedure pertaining to the porometer-fluorometer method is summarized in Figure 1.

## 2.4 | Data Analysis and Quality Control

$J$ in Equation (3) was calculated using the mean values of $\Phi_{PSII}$ and PPFD measured on the adaxial side illuminated by the sun (i.e., we did not use $\Phi_{PSII}$ and PPFD on the abaxial side: Figure 1). The $g_{tc}$ in Equation (5) was calculated using the mean values of $g_{bw}$ and $g_{sw}$ measured on each leaf side (Figure 1). $\Gamma^*$ and $R_d$ in Equations (6) and (7), respectively, were calculated using the mean values of $T_l$ from the abaxial side (Figure 1).

Although air temperature and relative humidity did not vary abruptly during each measurement period (lasting several minutes to tens of minutes), light conditions occasionally fluctuated between the measurements of the LI-6800 system and the LI-600 porometer because of the sudden cloud cover or sun beam. We excluded data recorded under such light conditions and finally used 310 samples for the following analysis. These samples included a wide range of environmental and physiological conditions (Supporting Information S1: Figure S1) and were considered sufficient for model construction and validation.

## 2.5 | Calibration and Validation for Estimating Leaf $CO_2$ Assimilation

To assess the accuracy and applicability of the porometer-fluorometer method for estimating $A_n$, cross-validation was performed while empirically calibrating the lumped parameter $s$ in Equation (3) using the following two approaches:

1. Constant $s$: Constant $s$ is valid under low-light conditions, particularly with artificial light (e.g., Loreto et al. 2009; van der Putten et al. 2018). Therefore, we firstly tested $A_n$ estimation using a constant $s$. First, all the data were divided into 14 datasets for each cultivar. Then, $s$ was calibrated on all the samples except for one data set, and $A_n$ was calculated for the left-out data set using the calibrated $s$. This process was repeated 14 times until all datasets were tested. The calibration of $s$ was performed by minimizing the root mean square error (RMSE) between the estimated and observed values of $A_n$ using differential evolution with the DEoptim package in R.
2. Variable $s$: McClain and Sharkey (2020) indicated that $s$ decreases with an increase in $J$. Therefore, we tested $A_n$ estimation using a variable $s$ as a function of $\Phi_{PSII} \times$ PPFD, which serves as a proxy for $J$. The same cross-validation described above was applied, whereas $s$ was calibrated using the following empirical equation:

$$s = \frac{a_s}{1 + \exp[b_s(\Phi_{PSII}\mathrm{PPFD} - c_s)]} + d_s, \quad (9)$$

where $a_s$, $b_s$, $c_s$, and $d_s$ are the calibrated parameters. This equation shows a decreasing sigmoid curve from $a_s + d_s$ (upper limit) to $d_s$ (lower limit). The parameter $b_s$ is related to the decreasing rate of s, and $c_s$ is the value of $\Phi_{PSII}$ PPFD when s is $a_s/2 + d_s$. These parameters were calibrated by minimizing the RMSE between the estimated and observed values of $A_n$ using the differential evolution described above. The R program for calibration and cross-validation is provided in Supporting Information S3.

These two validation methods were designed to assess the applicability of the common parameters across various species and cultivars. The use of common parameters minimizes the calibration frequency and enhances the efficiency of $A_n$ estimation.

# 3 | Results

$T_a$, $h$, and PPFD in the LI-6800 system were closely aligned with those in the LI-600 porometer, showing a minimal deviation in environmental conditions between the two instruments (Figure 2a–c). $T_l$ and leaf vapor pressure deficit ($VPD_l$), which strongly affect the physical and physiological processes of leaf photosynthesis, were also similar between the two instruments (Figure 2d,e). The $g_{tc}$ determined from the adaxial and abaxial values of $g_{sw}$ and $g_{bw}$ measured using the LI-600 porometer were similar to the both-sided $g_{tc}$ determined from the LI-6800 system, although an overestimation was observed at high $g_{tc}$ values above 0.5 mol $m^{-2}$ $s^{-1}$ (Figure 2f). Although the cause of this overestimation requires further investigation, this error has a limited impact on the $A_n$ estimation because $A_n$ is primarily constrained by other factors when $g_{tc}$ is significantly high (Supporting Information S1: Figure S2).

The relationships between $A_n$ and PPFD, $\Phi_{PSII}$PPFD, $g_{tc}$, and $T_l$ were nonlinear and species-specific (Figure 3). $A_n$ increased with PPFD at low PPFD, but became saturated at high PPFD, with saturation points varying by species (Figure 3a). Although $A_n$ increased with $\Phi_{PSII}$PPFD, the slope of this relationship varied across species (Figure 3b). $A_n$ increased with $g_{tc}$ at low $g_{tc}$ ($< 0.2$ mol $m^{-2}$ $s^{-1}$), but saturated at higher $g_{tc}$, with saturation points varying by species (Figure 3c). The relationship between $A_n$ and $T_l$ showed peaks at a species-specific optimal temperature (Figure 3d). These findings indicate that the individual variables measured by the LI-600 porometer and fluorometer alone cannot provide accurate $A_n$ estimates across different environments and species (Supporting Information S1: Figure S3).

The biochemical photosynthesis model combined with measurements using a porometer and a fluorometer substantially

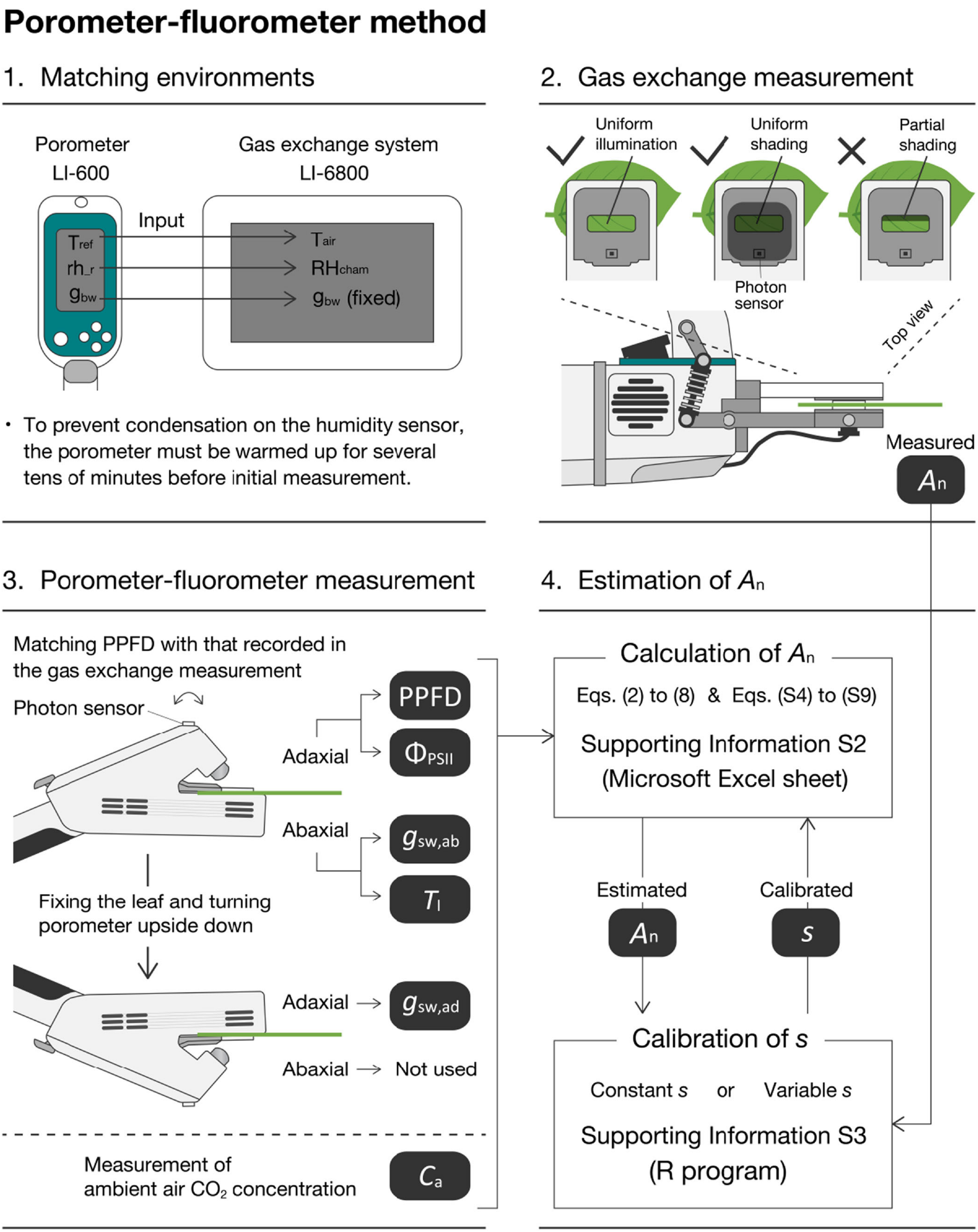


**FIGURE 1** | Procedure for estimating leaf $CO_2$ assimilation rate ($A_n$) using the porometer-fluorometer method. See Section 2 for detailed explanations.

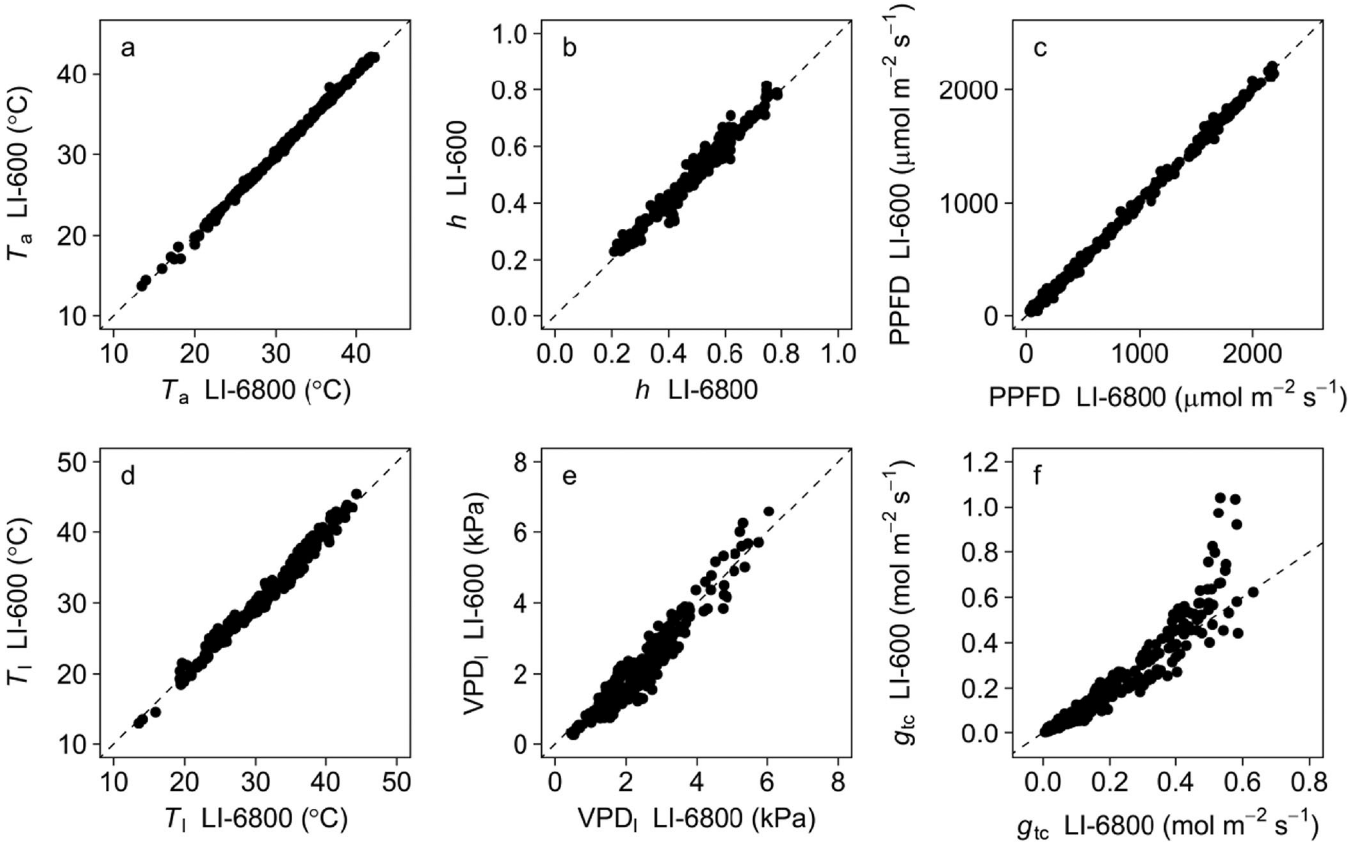


**FIGURE 2** | Comparison of (a) air temperature ($T_a$), (b) relative humidity ($h$), (c) photosynthetic photon flux density (PPFD), (d) leaf temperature ($T_l$), (e) leaf vapor pressure deficit (VPD$_l$), and (f) leaf total conductance for $CO_2$ transfer ($g_{tc}$) between the gas exchange measurement system (LI-6800) and the porometer equipped with the chlorophyll fluorometer (LI-600).

reduced the environment- and species-dependent biases in the estimation (Figure 4). However, $A_n$ was significantly overestimated when the parameter $s$ was fixed at 0.42 resulted from the often used values of $\alpha = 0.84$ and $\beta = 0.5$ (Figure 4). Furthermore, the degree of overestimation increased as $A_n$ increased (Figure 4), indicating an intensity-dependent change in $s$.

The calibration of $s$ reduced the overestimation of $A_n$ and reproduced the variations in the measured $A_n$ across diverse environments and species (Figure 5a–d). Calibration using the constant $s$ resulted in good estimates of $A_n$ (Figure 5a) although negative and positive error biases were observed, ranging from 5 to 15 μmol m$^{-2}$ s$^{-1}$ and exceeding 20 μmol m$^{-2}$ s$^{-1}$, respectively (Figure 5c). Calibration using the variable $s$ as a function of $\Phi_{PSII}$PPFD reduced the biases compared with the constant $s$ approach, providing more reasonable $A_n$ estimates (Figure 5b,d). For both constant and variable $s$, individual calibration for each species further improved estimation accuracy (Supporting Information S1: Figures S4 and S5), indicating species- and cultivar-dependent $s$ (Supporting Information S1: Table S1).

The calibrated value of constant $s$ for all species was 0.259, which was 38% lower than the often-used value (dashed line in Figure 6). The variable $s$ ranged from 0.286 to 0.246 as $\Phi_{PSII}$PPFD increased (solid line in Figure 6), 32%–41% lower than the often-used value. The large variations in individual $s$ values at low $\Phi_{PSII}$PPFD values, shown in Figure 6, were due to the uncertainty in $R_d$ when calculating $s$, rather than the actual variations in $s$. Equations (2) and (3) show the relative influence of $R_d$ on $s$, where it becomes more significant when $\Phi$PSII × PPFD is low.

## 4 | Discussion

### 4.1 | Accuracy of Estimating Leaf $CO_2$ Assimilation Using the Porometer-Fluorometer Method

The porometer-fluorometer method measures both $g_s$ and $\Phi_{PSII}$, enabling the consideration of both stomatal and non-stomatal photosynthesis limitations. Consequently, this method improves the accuracy of $A_n$ estimates compared to conventional empirical models that use only one of these variables (Figure 5 and Supporting Information S1: Figure S3). Additionally, the parameters of these empirical models show substantial species-specific variations and vary with growth environments, even within the same species. This increases errors in $A_n$ estimates when the empirical models are applied across a wide range of species and environmental conditions. In contrast, the calibrated parameter $s$ in the porometer-fluorometer method shows significantly smaller variations than the parameters of the empirical models (Supporting Information S1: Figure S3, Table S1), thereby reducing uncertainties in estimating $A_n$ under varying species and

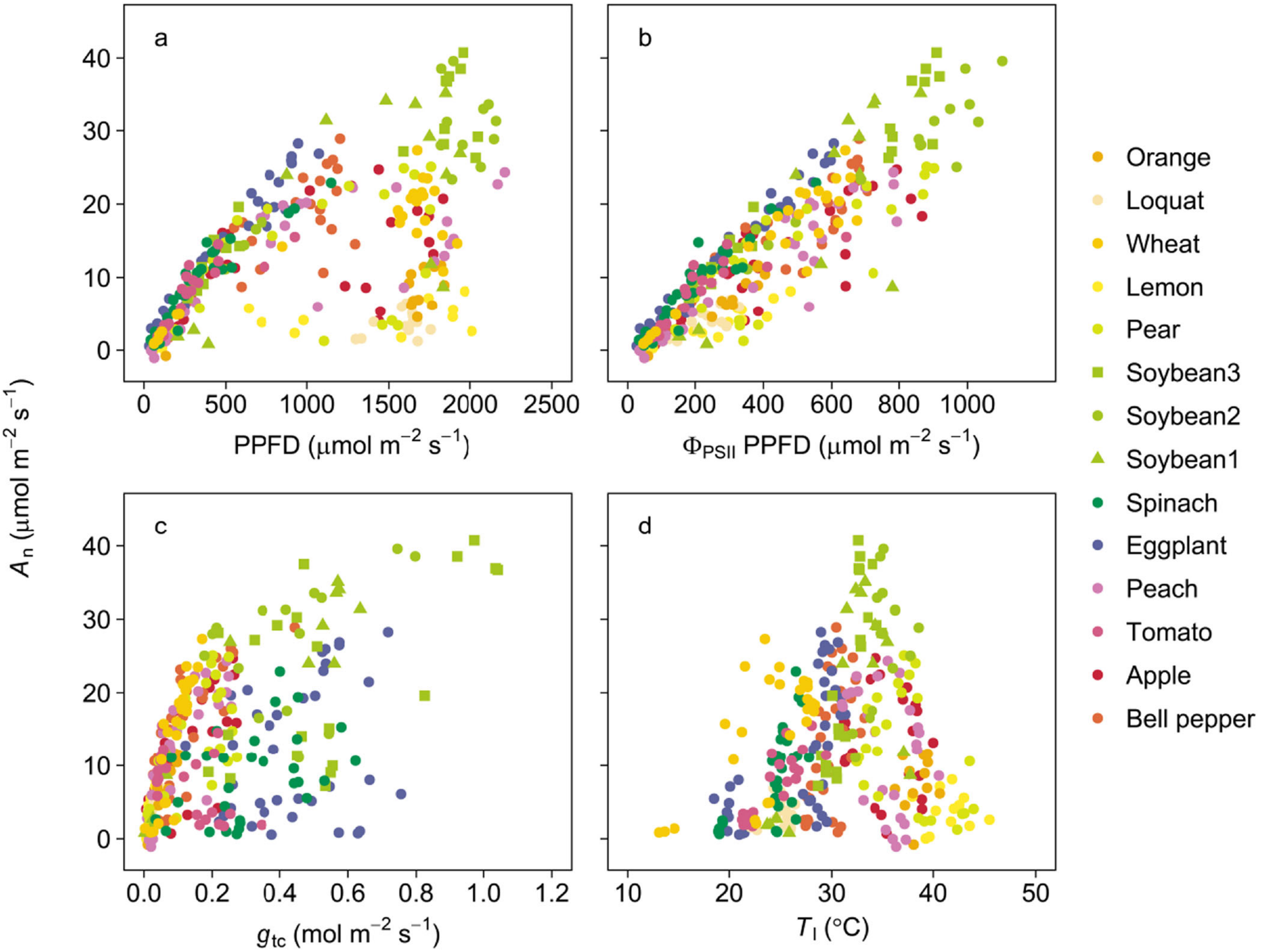


**FIGURE 3** | Relationship between leaf $CO_2$ assimilation rate ($A_n$) and (a) photosynthetic photon flux density (PPFD), (b) quantum yield of photochemistry in PSII ($\Phi_{PSII}$) × PPFD, which is the proxy of electron transport rate, (c) leaf total conductance for $CO_2$ transfer ($g_{tc}$), and (d) leaf temperature ($T_l$) in 14 cultivars among 12 species. $A_n$ was measured by the gas exchange measurement system (LI-6800), and $\Phi_{PSII}$, PPFD, $g_{tc}$, and $T_l$ were inferred from the porometer equipped with the chlorophyll fluorometer (LI-600). [Color figure can be viewed at wileyonlinelibrary.com]

environmental conditions. In summary, the porometer-fluorometer method enables accurate $A_n$ estimation in diverse $C_3$ plants compared to conventional empirical approaches, provided appropriate calibration is performed.

In this study, the porometer-fluorometer method was validated under relatively stable environmental conditions. However, its accuracy under fluctuating environments, particularly fluctuating light conditions that cause highly variable and delayed photosynthetic response (i.e., photosynthetic induction), in addition to fluctuating VPD conditions that may cause decline in $g_s$ and $A_n$ (Inoue et al. 2021), remains unclear. Short measurement intervals of variables can reduce the estimation error of $A_n$ due to photosynthetic induction even when assuming a steady-state (Murakami and Jishi 2022). Using a "Manual Mode" of the LI-600 porometer, which allows continuous recording at intervals of 1–2 s while keeping the leaf clamped, may address this requirement. However, further investigation is required to determine the acceptable measurement interval and whether such an interval is feasible with a porometer and fluorometer.

### 4.2 | Remaining Uncertainties in the Calibrated Parameter *s* and Possible Ideas to Reduce Them

The porometer-fluorometer method effectively estimates $A_n$ across diverse species, cultivars, and environments; however, uncertainties in the calibrated parameter $s$, require further investigation for more accurate estimation of $A_n$. Previous studies indicate that $s$ may be consistently lower than the often-used value of 0.42. Loreto et al. (2009) found that $s$ was 35–45% lower under artificial light with a high blue-to-red ratio, whereas McClain and Sharkey (2020) observed a 27%–31% decrease in $s$, likely due to blue light absorption by non-photosynthetic pigments, such as carotenoids. Yin et al. (2009) estimated $s$ to be 5%–20% lower than the often-used values, suggesting alternative electron transport pathways such as cyclic electron flow. Another source of lower $s$ values is $\Phi_{PSII}$ overvaluations caused by mesophyll layer heterogeneity within the leaf. Evans (2009) demonstrated that mesophyll layers have different $\Phi_{PSII}$ because of different photochemical properties and amount of absorbed light in each layer, causing overvaluations when measuring $\Phi_{PSII}$ on one side of leaf surface.

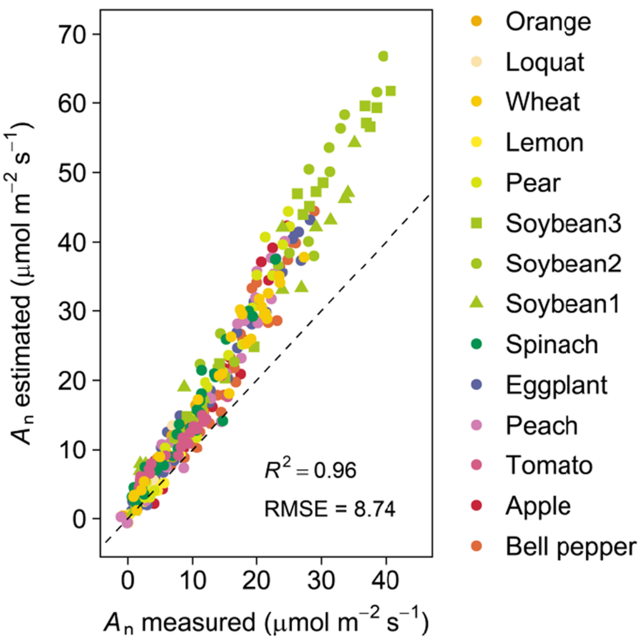


**FIGURE 4** | Relationship between estimated and measured leaf $CO_2$ assimilation rate ($A_n$) in 14 cultivars belonging to 12 species. $A_n$ was estimated with the parameter $s$ in Equation (3) fixing at the often-used value of 0.42. The dashed line represents 1 to 1 line. [Color figure can be viewed at wileyonlinelibrary.com]

These overvaluations increased under actinic light with high blue-to-red ratios, likely owing to high blue absorption in the top layers of the leaf and concomitant large heterogeneity (Evans et al. 2017). Moreover, Evans (2009) reported that heterogeneity increased with increasing light intensity, and these intensity-dependent changes were large under higher blue-to-red light ratios (Evans et al. 2017). Natural sunlight contains higher blue fractions (approximately 50%: Kotilainen et al. 2020) than commonly used artificial lights (such as 10% blue). Therefore, the low values and intensity-dependent changes in $s$ observed in this study were likely driven by non-photosynthetic light absorption and/or mesophyll layer heterogeneity, with alternative electron transport pathways playing a small role. Additionally, the present study found that woody plants, which tend to have thicker leaves and thus greater mesophyll layer heterogeneity than herbaceous species (Borsuk and Brodersen 2019), exhibited more pronounced light intensity-dependent changes in $s$ (Supporting Information S1: Figure S4), supporting the hypothesis that overvaluations of $\Phi_{PSII}$ owing to mesophyll layer heterogeneity affected variability in $s$. However, this study could not independently assess the impact of each factor, highlighting the need for further investigation under sunlight conditions.

Sunlight spectrum changes diurnally and seasonally with change in solar elevation angles and atmospheric conditions. Therefore, it is reasonable to assume that the value of $s$ also varies over time. In the present study, however, even a constant value of $s$ enabled accurate estimation of $A_n$ although validation was performed across a range of seasons and times of day, suggesting a limited effect of temporal variation in $s$ on $A_n$ estimates. This is likely because diurnal and seasonal variations of blue-to-red ratios of sunlight are generally modest under many conditions (typically approximately 45%–55% at solar elevation angles > 20°: Kotilainen et al. 2020). Nevertheless, under conditions where the blue-to-red light ratio changes dramatically (at solar elevation angles < 20°), temporal variability in $s$ may become more significant.

Uncertainties in $s$ described above could be reduced using some approaches. The effects of non-photosynthetic light absorption can be corrected using relative blue efficiency (ratio of quantum yield for blue and red light) (McClain and Sharkey 2020). Directly measuring mesophyll layer heterogeneity in the field remains challenging; however, a possible approach is to estimate it empirically using the leaf adaxial/abaxial chlorophyll fluorescence ratio, which is related to the intra-leaf light absorption gradient (Mand et al. 2013). This ratio can be easily obtained using a fluorometer by turning the leaf over without additional instruments. Additionally, machine learning methods based on leaf hyperspectral reflectance can detect chloroplast movements (Hermanowicz and Łabuz 2024), potentially correcting for mesophyll heterogeneity effects varying with light conditions.

Uncertainties in conventional model parameters also affect the value of $s$ in $A_n$ estimation. Typically, $\alpha$ is assumed to be approximately 0.85, and this assumption is reasonable for $A_n$ estimates in most healthy leaves (Supporting Information S1: Figure S6). However, $\alpha$ should be carefully determined for senescent or drought-stressed leaves, as it may be significantly lower (Gates 1980; Ehleringer 1980). $\alpha$ values depend on leaf chlorophyll content (Gabrielsen 1948) and thus, can be individually estimated from spectral indices to retrieve leaf chlorophyll content (e.g., the present study used NDVIgreen for estimating $\alpha$ in 12 species; Supporting Information S1: Figure S6). Often, $\beta$ is assumed to be 0.5, but its value varies under far-red light conditions (Wientjes et al. 2017; Murakami et al. 2018). However, individual determination of $\beta$ is challenging under field conditions. Although $\Gamma^*$ value is often taken from a literature source, it generally varies among species (Hermida-Carrera et al. 2016). The present study predicted that an extremely overestimated or underestimated $\Gamma^*$ value would increase the error in $A_n$ estimate, although a dire error in $A_n$ could be avoided (Supporting Information S1: Figure S7). Individual determination of $\Gamma^*$ can provide more reasonable $A_n$ estimates, but it remains challenging, as well as in the case of $\beta$. In the present study, $g_m$ was assumed to be infinite, and the effect of $g_m$ on $A_n$ was partially included in the calibrated values of $s$. This assumption is useful when $g_m$ is independently high or comparable to or higher than $g_{sc}$, as is likely the case for most of the data in this study (Supporting Information S1: Figure S8). However, assuming infinite $g_m$ does not hold under certain conditions. When $g_m$ coordinates with $g_{sc}$, estimation error in $A_n$ increases when an absolute value of $g_m$ is low and its value is lower than that of $g_{sc}$ (Supporting Information S1: Figure S8). These errors cannot be fully corrected using only a single parameter $s$ (Supporting Information S1: Figure S8). Therefore, in leaves under such extreme conditions (e.g., severe drought), individual determination or calibration of $g_m$ is recommended for a more accurate estimation of $A_n$ (Niinemets et al. 2009). Uncertainties in $R_d$ are more problematic than those in other parameters because $R_d$ is the intercept in Equation (2), and parameter $s$, which is the slope in Equation (2), cannot fully

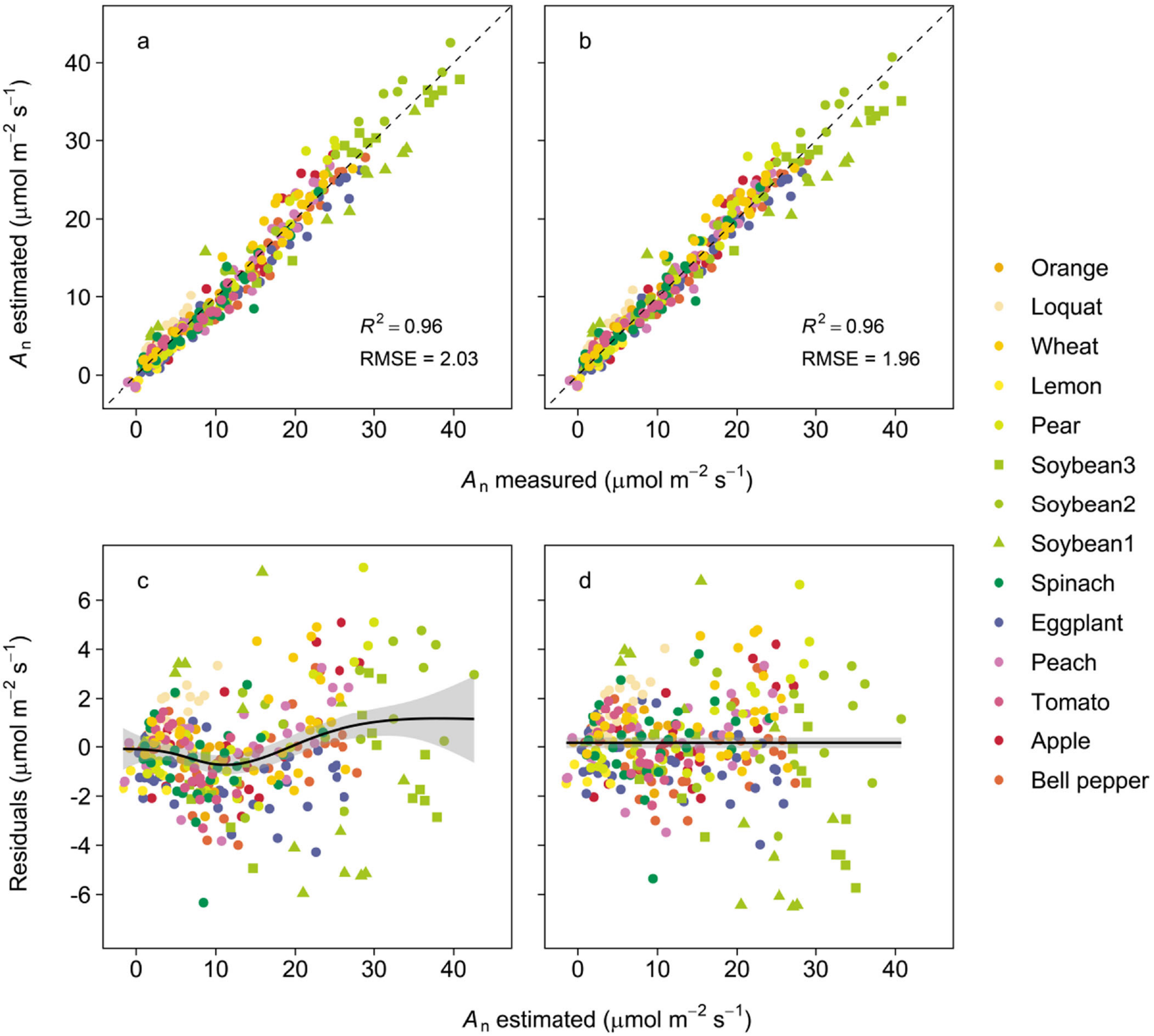


**FIGURE 5** | Relationship between estimated and measured leaf $CO_2$ assimilation rate ($A_n$) in 14 cultivars belonging to 12 species. $A_n$ was estimated using parameter *s* in Equation (3) calibrating under the assumption that (a and c) *s* is constant and (b and d) *s* varies with quantum yield of photochemistry in PSII ($\Phi_{PSII}$) × photosynthetic photon flux density (PPFD). In (a and b), the results of cross-validation are shown, and the dashed lines represent 1 to 1 line. In (c and d), residuals of $A_n$ and their biased lines fitted using a generalized additive model are shown. [Color figure can be viewed at wileyonlinelibrary.com]

calibrate the effect of $R_d$. In particular, $R_d$ should be carefully determined at a low $A_n$, where the relative contribution of $R_d$ is large (Supporting Information S1: Figure S9). $g_m$ and $R_d$ can be simultaneously estimated using the non-rectangular hyperbolic model combined with measurements of gas exchange and chlorophyll fluorescence under photorespiratory conditions (i.e., under ambient air conditions) (Yin and Struik 2009; Yin and Amthor 2024), although this method requires step changes in $CO_2$ and PPFD.

Despite these uncertainties, the proposed method reproduced $A_n$ variations with a common parameter set across species. This may be due to reduced parameter variations through coordination within the photosynthetic system between electron transport and carbon metabolism. Assuming that the energy distribution in photosystems and cyclic and pseudo-cyclic electron transports are coordinated with linear electron transport to fulfill carbon metabolism demands (Allen 2003; Johnson and Berry 2021), their effects (i.e., $\beta$, $f_{pseudo}$, $f_{cyc}$, and resultant *s* in Equation [3]) are functions of $\Phi_{PSII}$, as modeled in this study. Thus, variations in $\beta$, $f_{pseudo}$, and $f_{cyc}$ are partly reflected in the measured $\Phi_{PSII}$, leading to smaller variations in the calibrated parameter *s* compared to scenarios without coordination. However, such coordination is nonlinear, and there are limits to imposing all uncertainties on one parameter. Ideally, these parameters discussed in this section should be determined individually to reduce the uncertainties in *s*. However, individual estimations are currently time-consuming or technically

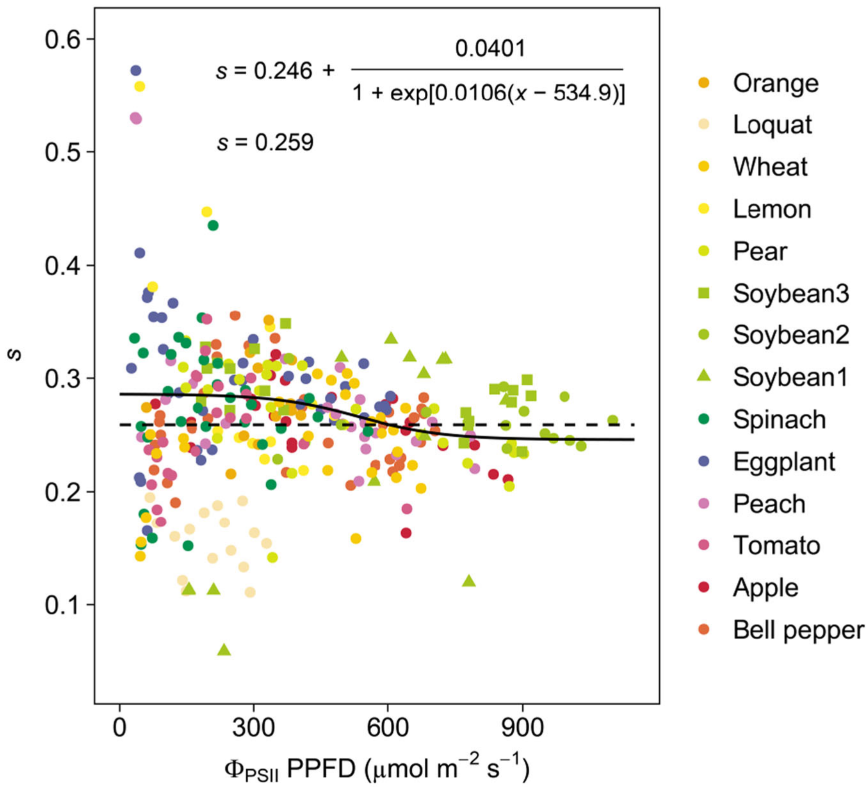


**FIGURE 6** | Responses of calibrated parameter $s$ in Equation (3) to quantum yield of photochemistry in PSII ($\Phi_{PSII}$) × photosynthetic photon flux density (PPFD). $s$ was calculated from Equations (2) and (3) by substituting the measured values of $A_n$, $C_i$, $\Phi_{PSII}$, and PPFD and the estimated values of $\Gamma^*$ and $R_d$ from Equations (6)–(8). The regression lines and fitted equations under the assumption that $s$ is constant (dashed line) and that $s$ varies with $\Phi_{PSII}$PPFD (solid line) are also shown. [Color figure can be viewed at wileyonlinelibrary.com]

difficult under field conditions, which conflicts with the goal of high-throughput estimation of $A_n$. The proposed method, which incorporates some uncertainties into the parameter $s$, offers a temporary solution that balances the accuracy, applicability, and efficiency in estimating $A_n$ in the field.

## 4.3 | Advantages and Disadvantages of the Porometer-Fluorometer Method

### 4.3.1 | Versus Gas Exchange Measurements

The compact, lightweight design of the porometer with the chlorophyll fluorometer enables on-the-go measurements in the field, and its fast stabilization time allows researchers to obtain high spatiotemporal resolution data on $A_n$. This setup is more advantageous for high-throughput field phenotyping of photosynthesis than gas exchange measurements. While using multiple gas exchange systems can also perform high-throughput phenotyping, only a limited number of researchers can afford this approach because of its cost and the manpower required. The porometer-fluorometer method offers a cost-effective and labor-efficient alternative, making it accessible to a broad range of researchers. Moreover, the simple process of clamping leaves in addition to the compact design may enable the automatic estimation of $A_n$ using techniques for harvesting robots developed in greenhouse agriculture (Zhao et al. 2016). For example, a robot that can automatically detect and clamp leaves with a porometer and chlorophyll fluorometer mounted on its arm can be expected. This further reduces labor costs and increases the efficiency of collecting data of $A_n$.

The porometer-fluorometer method requires empirical calibration due to uncertainties in plant physiological responses, whereas gas exchange measurements are primarily based on the physical process of leaf gas exchange and can independently measure $A_n$ from the physiological mechanisms. Therefore, the accuracy of $A_n$ evaluation in the porometer-fluorometer method is lower than that of gas exchange measurements, and for precise $A_n$ evaluation, the conventional gas exchange system remains the gold standard. The porometer-fluorometer method may be suitable for identifying leaves and cultivars with relatively high $A_n$ for screening (such as breeding programs in agriculture) but less so for detailed studies of specific photosynthetic processes where accuracy is paramount.

### 4.3.2 | Versus Proximal and Remote Sensing Techniques

The porometer-fluorometer method provides contact measurement of $\Phi_{PSII}$ and $g_s$ to estimate $A_n$. In contrast, recent advances in light-induce fluorescence transient (LIFT) method (Kolber et al. 1998; Keller et al. 2019) and sun-induced chlorophyll fluorescence (SIF) measurement (Porcar-Castell et al. 2021) have enabled proximal and remote sensing of $\Phi_{PSII}$ (or related variables) across various scales. Additionally, leaf temperature measurements combined with energy balance analysis enable proximal and remote sensing of $g_s$ (Jones 1999; Vialet-Chabrand and Lawson 2019). These techniques can be also combined with a biochemical photosynthesis model to estimate $A_n$ (Kimura et al. 2024), as performed in this study. Unlike the porometer-fluorometer method, these techniques not only allow for the noncontact estimation of $A_n$ but also enable the instant detection of spatial distributions of $A_n$ when combined with imaging techniques. Therefore, throughput of the porometer-fluorometer method may be lower than that of proximal and remote sensing approaches. However, instruments simultaneously detecting $\Phi_{PSII}$ and $g_s$ for proximal and remote sensing have not yet been commercialized, and thus the porometer-fluorometer method is currently the most accessible for most researchers.

The porometer-fluorometer method measures $g_s$ and $\Phi_{PSII}$ directly, reducing uncertainties in $A_n$ estimates compared with relying on empirical estimates of $g_s$ and $\Phi_{PSII}$. In contrast, proximal and remote sensing approaches require additional models to estimate $J$ from SIF (Gu et al. 2019; Han et al. 2022) and supplementary measurements using reference materials to estimate $g_s$ from thermal imaging (Jones 1999; Vialet-Chabrand and Lawson 2019), increasing uncertainties in $A_n$ estimates. Therefore, $A_n$ estimate accuracy in the porometer-fluorometer method is generally higher than that when using proximal and remote sensing approaches.

#### 4.3.3 | Combination With the Conventional Methods

Disadvantages of the porometer-fluorometer method can be compensated in combination with gas exchange measurements and proximal and remote sensing techniques. The porometer-fluorometer method can complement gas exchange systems, offering high-throughput sampling across a large field with precise calibration of small representative samples. Moreover, this method can be used to calibrate $g_s$ and $\Phi_{PSII}$ estimated from proximal and remote sensing, further increasing the throughput of $A_n$ estimates with maintained accuracy. Overall, the porometer-fluorometer method is not necessarily a replacement for existing methods and is valuable when used in combination with other methods.


### Acknowledgements

This study was partially supported by the Japan Society for the Promotion of Science (JSPS) KAKENHI, under Grant Number 21K14950 and 25H00953. This study was also supported by a Cabinet Office grant in aid, the Advanced Next-Generation Greenhouse Horticulture by IoP (Internet of Plants), Japan, and by a Cabinet Office grant in aid, Evolution to Society 5.0 Agriculture Driven by IoP (Internet of Plants), Japan. We thank Toyoshi Omizu, Takahiro Mikuni, Akihito Okubo, Koji Hashimoto, Rina Suzuki, Yuki Muta, Kanae Saito, Kazumi Suenaga, Ayumi Hatakeyama, Hiroko Yamaura and Kohei Nishiya (National Agriculture and Food Research Organization) for their technical assistance.


### Conflicts of Interest

The authors declare no conflicts of interest.

### Data Availability Statement

The data used to reproduce the results of this study are available in Supporting Information S2 and S3. Supporting Information S2: Microsoft Excel sheet including data and formulae for calculating An. Supporting Information S3: R program for calibrating the parameter s and performing cross-validation, and Supporting Information S2 can be read into this program as a template. Directions for their usage are described in the files.

### References


Allen, J. F. 2003. "Cyclic, Pseudocyclic and Noncyclic Photophosphorylation: New Links in the Chain." *Trends in Plant Science* 8: 15–19.

Borsuk, A. M., and C. R. Brodersen. 2019. "The Spatial Distribution of Chlorophyll in Leaves." *Plant Physiology* 180: 1406–1417.

Chow, W. S., A. Melis, and J. M. Anderson. 1990. "Adjustments of Photosystem Stoichiometry in Chloroplasts Improve the Quantum Efficiency of Photosynthesis." *Proceedings of the National Academy of Sciences* 87: 7502–7506.

Ehleringer, J. 1980. "Leaf Morphology and Reflectance in Relation to Water and Temperature Stress." In *Adaptations of Plants to Water and High Temperature Stress*, edited by N. C. Turner and P. J. Kramer, 295–308. Wiley.

Evans, J. R. 2009. "Potential Errors in Electron Transport Rates Calculated From Chlorophyll Fluorescence as Revealed by a Multilayer Leaf Model." *Plant and Cell Physiology* 50: 698–706.

Evans, J. R., P. B. Morgan, and S. Von Caemmerer. 2017. "Light Quality Affects Chloroplast Electron Transport Rates Estimated From Chl Fluorescence Measurements." *Plant and Cell Physiology* 58: 1652–1660.

Fang, L., X. Yin, P. E. L. van der Putten, P. Martre, and P. C. Struik. 2022. "Drought Exerts a Greater Influence Than Growth Temperature on the Temperature Response of Leaf Day Respiration in Wheat (*Triticum aestivum*)." *Plant, Cell and Environment* 45: 2062–2077.

Farquhar, G. D., S. von Caemmerer, and J. A. Berry. 1980. "A Biochemical Model of Photosynthetic $CO_2$ Assimilation in Leaves of $C_3$ Species." *Planta* 149: 78–90.

Gabrielsen, E. K. 1948. "Effects of Different Chlorophyll Concentrations on Photosynthesis in Foliage Leaves." *Physiologia Plantarum* 1: 5–37.

Galmés, J., C. Hermida-Carrera, L. Laanisto, and Ü. Niinemets. 2016. "A Compendium of Temperature Responses of Rubisco Kinetic Traits: Variability Among and Within Photosynthetic Groups and Impacts on Photosynthesis Modeling." *Journal of Experimental Botany* 67: 5067–5091.

Gates, D. M. 1980. *Biophysical Ecology*, 215–238. Springer-Verlag.

Genty, B., J. M. Briantais, and N. R. Baker. 1989. "The Relationship Between the Quantum Yield of Photosynthetic Electron Transport and Quenching of Chlorophyll Fluorescence." *Biochimica et Biophysica Acta (BBA) - General Subjects* 990: 87–92.

Gu, L., J. Han, J. D. Wood, C. Y. Y. Chang, and Y. Sun. 2019. "Sun-Induced Chl Fluorescence and its Importance for Biophysical Modeling of Photosynthesis Based on Light Reactions." *New Phytologist* 223: 1179–1191.

Han, J., C. Y. Y. Chang, L. Gu, et al. 2022. "The Physiological Basis for Estimating Photosynthesis From Chl a Fluorescence." *New Phytologist* 234: 1206–1219.

Harley, P. C., F. Loreto, G. Di Marco, and T. D. Sharkey. 1992. "Theoretical Considerations When Estimating the Mesophyll Conductance to $CO_2$ Flux by Analysis of the Response of Photosynthesis to $CO_2$." *Plant Physiology* 98: 1429–1436.

He, D., and G. E. Edwards. 1996. "Evaluation of the Potential to Measure Photosynthetic Rates in $C_3$ Plants (*Flaveria pringlei* and *Oryza sativa*) by Combining Chlorophyll Fluorescence Analysis and a Stomatal Conductance Model." *Plant, Cell and Environment* 19: 1272–1280.

Hermanowicz, P., and J. Łabuz. 2024. "Hyperspectral Imaging for Chloroplast Movement Detection." *Journal of Experimental Botany* 76, no. 3: 882–898.

Hermida-Carrera, C., M. V. Kapralov, and J. Galmés. 2016. "Rubisco Catalytic Properties and Temperature Response in Crops." *Plant Physiology* 171: 2549–2561.

Inoue, T., M. Sunaga, M. Ito, et al. 2021. "Minimizing VPD Fluctuations Maintains Higher Stomatal Conductance and Photosynthesis, Resulting in Improvement of Plant Growth in Lettuce." *Frontiers in Plant Science* 12: 646144.

Johnson, J. E., and J. A. Berry. 2021. "The Role of Cytochrome b6f in the Control of Steady-State Photosynthesis: A Conceptual and Quantitative Model." *Photosynthesis Research* 148: 101–136.

Jones, H. G. 1999. "Use of Thermography for Quantitative Studies of Spatial and Temporal Variation of Stomatal Conductance over Leaf Surfaces." *Plant, Cell and Environment* 22: 1043–1055.

Keller, B., I. Vass, S. Matsubara, et al. 2019. "Maximum Fluorescence and Electron Transport Kinetics Determined by Light-Induced Fluorescence Transients (LIFT) for Photosynthesis Phenotyping." *Photosynthesis Research* 140: 221–233.

Kimura, K., E. Kumagai, E. Fushimi, and A. Maruyama. 2024. "Alternative Method for Determining Leaf CO2 Assimilation Without Gas Exchange Measurements: Performance, Comparison and Sensitivity Analysis." *Plant, Cell and Environment* 47: 992–1002.

Kitao, M., Y. Yasuda, E. Kodani, et al. 2021. "Integration of Electron Flow Partitioning Improves Estimation of Photosynthetic Rate under Various Environmental Conditions Based on Chlorophyll Fluorescence." *Remote Sensing of Environment* 254: 112273.

Kolber, Z. S., O. Prášil, and P. G. Falkowski. 1998. "Measurements of Variable Chlorophyll Fluorescence Using Fast Repetition Rate Techniques: Defining Methodology and Experimental Protocols." *Biochimica et Biophysica Acta (BBA) - Bioenergetics* 1367: 88–106.

Kotilainen, T., P. Aphalo, C. Brelsford, et al. 2020. "Patterns in the Spectral Composition of Sunlight and Biologically Meaningful Spectral Photon Ratios as Affected by Atmospheric Factors." *Agricultural and Forest Meteorology* 291: 108041.

Laisk, A., and F. Loreto. 1996. "Determining Photosynthetic Parameters From Leaf CO2 Exchange and Chlorophyll Fluorescence (Ribulose-1,5-Bisphosphate Carboxylase/Oxygenase Specificity Factor, Dark Respiration in the Light, Excitation Distribution Between Photosystems, Alternative Electron Transport Rate, and Mesophyll Diffusion Resistance." *Plant Physiology* 110: 903–912.

Leuning, R. 1995. "A Critical Appraisal of Combine Stomatal Model C3 Plants." *Plant, Cell and Environment* 18: 339–355.

Loreto, F., T. Tsonev, and M. Centritto. 2009. "The Impact of Blue Light on Leaf Mesophyll Conductance." *Journal of Experimental Botany* 60: 2283–2290.

Loriaux, S. D., T. J. Avenson, J. M. Welles, et al. 2013. "Closing in on Maximum Yield of Chlorophyll Fluorescence Using a Single Multiphase Flash of Sub-Saturating Intensity." *Plant, Cell and Environment* 36: 1755–1770.

Mand, P., L. Hallik, J. Penuelas, and O. Kull. 2013. "Electron Transport Efficiency At Opposite Leaf Sides: Effect of Vertical Distribution of Leaf Angle, Structure, Chlorophyll Content and Species in a Forest Canopy." *Tree Physiology* 33: 202–210.

McClain, A. M., and T. D. Sharkey. 2020. "Building a Better Equation for Electron Transport Estimated From Chl Fluorescence." *New Phytologist* 225: 604–608.

Murakami, K., and T. Jishi. 2022. "Appropriate Time Interval of PPFD Measurement to Estimate Daily Photosynthetic Gain." *Functional Plant Biology* 49: 452–462.

Murakami, K., R. Matsuda, and K. Fujiwara. 2018. "Quantification of Excitation Energy Distribution Between Photosystems Based on a Mechanistic Model of Photosynthetic Electron Transport." *Plant, Cell and Environment* 41: 148–159.

Niinemets, Ü., A. Díaz-Espejo, J. Flexas, J. Galmés, and C. R. Warren. 2009. "Importance of Mesophyll Diffusion Conductance in Estimation of Plant Photosynthesis in the Field." *Journal of Experimental Botany* 60: 2271–2282.

Porcar-Castell, A., Z. Malenovský, T. Magney, et al. 2021. "Chlorophyll a Fluorescence Illuminates a Path Connecting Plant Molecular Biology to Earth-System Science." *Nature Plants* 7: 998–1009.

van der Putten, P. E. L., X. Yin, and P. C. Struik. 2018. "Calibration Matters: On the Procedure of Using the Chlorophyll Fluorescence Method to Estimate Mesophyll Conductance." *Journal of Plant Physiology* 220: 167–172.

Saathoff, A. J., and J. Welles. 2021. "Gas Exchange Measurements in the Unsteady State." *Plant, Cell and Environment* 44: 3509–3523.

Sharkey, T. D. 1985. "Photosynthesis in Intact Leaves of C3 Plants: Physics, Physiology and Rate Limitations." *Botanical Review* 51: 53–105.

Simkin, A. J., P. E. López-Calcagno, and C. A. Raines. 2019. "Feeding the World: Improving Photosynthetic Efficiency for Sustainable Crop Production." *Journal of Experimental Botany* 70: 1119–1140.

Vialet-Chabrand, S., and T. Lawson. 2019. "Dynamic Leaf Energy Balance: Deriving Stomatal Conductance From Thermal Imaging in a Dynamic Environment." *Journal of Experimental Botany* 70: 2839–2855.

Wientjes, E., J. Philippi, J. W. Borst, and H. van Amerongen. 2017. "Imaging the Photosystem I/Photosystem II Chlorophyll Ratio Inside the Leaf." *Biochimica et Biophysica Acta (BBA)–Bioenergetics* 1858: 259–265.

Yin, X., and J. S. Amthor. 2024. "Estimating Leaf Day Respiration From Conventional Gas Exchange Measurements." *New Phytologist* 241: 52–58.

Yin, X., and P. C. Struik. 2009. "Theoretical Reconsiderations When Estimating the Mesophyll Conductance to $CO_2$ Diffusion In Leaves of $C_3$ Plants by Analysis of Combined Gas Exchange and Chlorophyll Fluorescence Measurements." *Plant, Cell and Environment* 32: 1513–1524.

Yin, X., P. C. Struik, P. Romero, et al. 2009. "Using Combined Measurements of Gas Exchange and Chlorophyll Fluorescence to Estimate Parameters of a Biochemical C Photosynthesis Model: A Critical Appraisal and a New Integrated Approach Applied to Leaves in a Wheat (*Triticum aestivum*) Canopy." *Plant, Cell and Environment* 32: 448–464.

Zhao, Y., L. Gong, Y. Huang, and C. Liu. 2016. "A Review of Key Techniques of Vision-Based Control for Harvesting Robot." *Computers and Electronics in Agriculture* 127: 311–323.

Zhu, X. G., S. P. Long, and D. R. Ort. 2010. "Improving Photosynthetic Efficiency for Greater Yield." *Annual Review of Plant Biology* 61: 235–261.

## Supporting Information

Additional supporting information can be found online in the Supporting Information section.